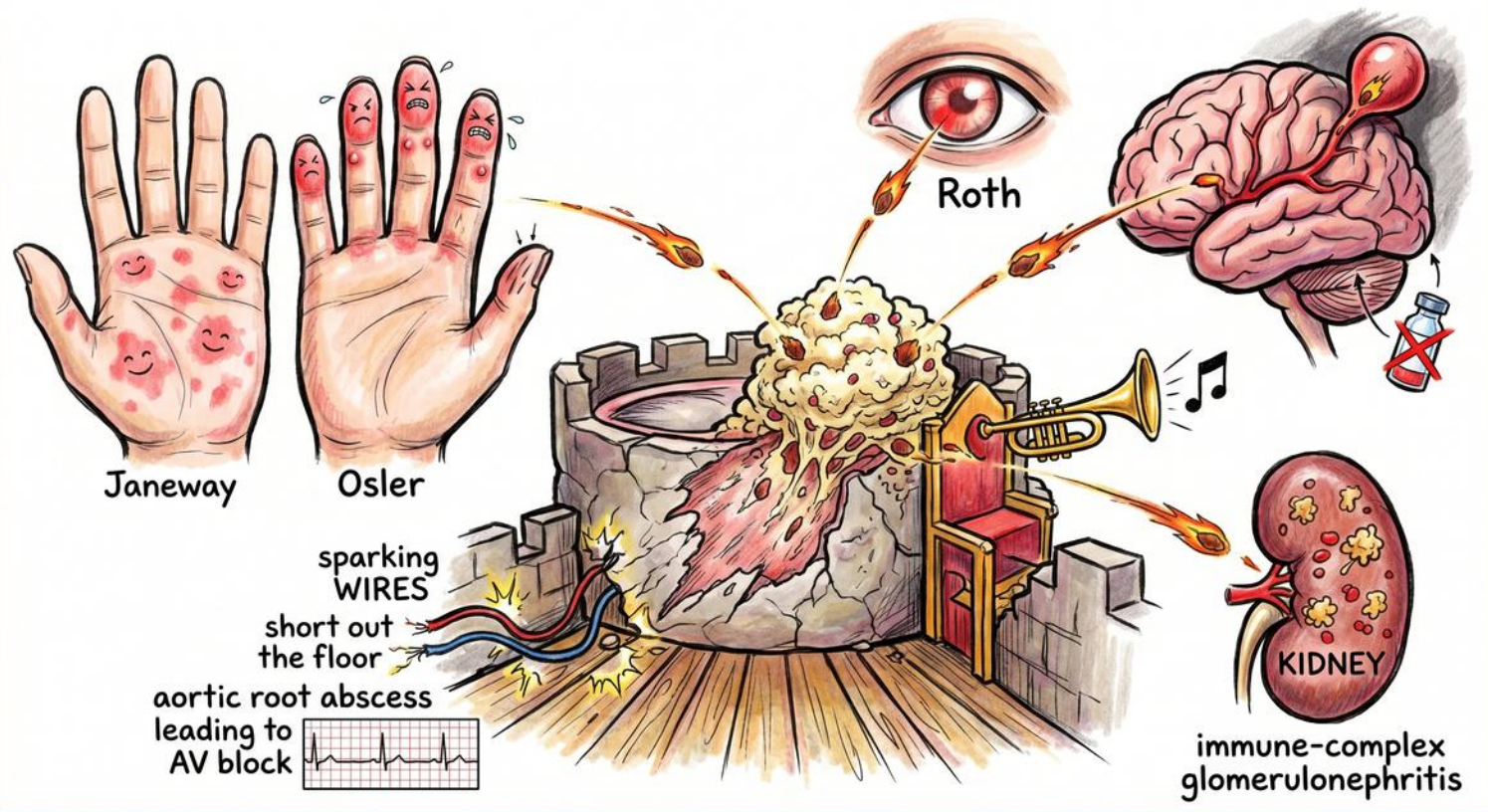


Infective Endocarditis — Clinical & Complications





Vegetation throwing emboli

WHAT YOU SEE

The crumbling valve-throne with a cauliflower vegetation firing flaming arrows outward is the source showering septic emboli around the body.

WHAT IT ENCODES

The infected vegetation continuously sheds fragments that embolize. Highest embolic risk: vegetation >10 mm AND mobile, *S. aureus*, MITRAL (especially anterior leaflet) location, and prior embolism. Embolic risk peaks in the first 1–2 weeks and falls rapidly once effective antibiotics start, so a large mobile vegetation that has already embolized (or is >10 mm with another surgical indication) argues for early surgery.

THE BOARD TRAP

'Add aspirin/anticoagulation to prevent embolism' -> wrong; antiplatelet/anticoagulant therapy does NOT reduce embolic risk in IE and increases bleeding — the large mobile vegetation is itself a surgical indication, so the answer is early surgery, not anticoagulation.



Janeway lesions (painless palms)

WHAT YOU SEE

The palm dotted with flat, smiling, painLESS red macules is Janeway lesions from septic microemboli.

WHAT IT ENCODES

JANEWAY lesions = painLESS, flat, erythematous macules on the PALMS and SOLES from septic microemboli — a VASCULAR phenomenon and a minor Duke criterion. The smiling faces in the sketch cue 'no pain.'

THE BOARD TRAP

'Janeway lesions are painful and immune-mediated' -> wrong; Janeway are painLESS, flat, VASCULAR (embolic) lesions on palms/soles — the painful immune ones are Osler nodes.



Osler nodes (painful pads)

WHAT YOU SEE

The fingers with raised, tender, grimacing nodules on the pads are Osler nodes from immune-complex deposition.

WHAT IT ENCODES

OSLER nodes = painful, raised, tender nodules on the PADS of fingers/toes from immune-complex deposition — an IMMUNOLOGIC phenomenon and a minor Duke criterion, typical of subacute IE. Splinter hemorrhages (linear subungual streaks) are a related microembolic finding. Mnemonic: 'Osler = Ouch.'

THE BOARD TRAP

'Osler nodes are painless vascular emboli' -> wrong; Osler nodes are PAINFUL and IMMUNOLOGIC (immune complexes) on the finger/toe pads — the painless embolic macules on palms/soles are Janeway lesions.



Splinter hemorrhages (fingertips)

WHAT YOU SEE

The dark linear streaks under the fingernails on the Osler-node hand are splinter hemorrhages from nailbed microemboli.

WHAT IT ENCODES

Splinter hemorrhages are linear, reddish-brown subungual streaks caused by microemboli (or immune microvascular injury) to the nailbed capillaries, one of the peripheral stigmata of IE. Like Janeway and Osler findings they are supportive, nonspecific signs — also seen with trauma — and feed the vascular/immunologic minor Duke categories rather than being diagnostic alone.



Septic emboli to spleen/systemic

WHAT YOU SEE

The flaming arrows fanning out from the vegetation toward the body's organs represent left-sided septic emboli to spleen, kidneys, and limbs.

WHAT IT ENCODES

Left-sided vegetations embolize systemically. SPLEEN: septic emboli cause infarcts (LUQ pain) and rarely a splenic abscess — persistent fever/bacteremia despite therapy warrants imaging and drainage. Renal septic emboli cause flank pain/infarcts (distinct from the immune-complex GN), and limb/mesenteric emboli can occur. The brain is the single most common systemic target.



Roth spots (eye)

WHAT YOU SEE

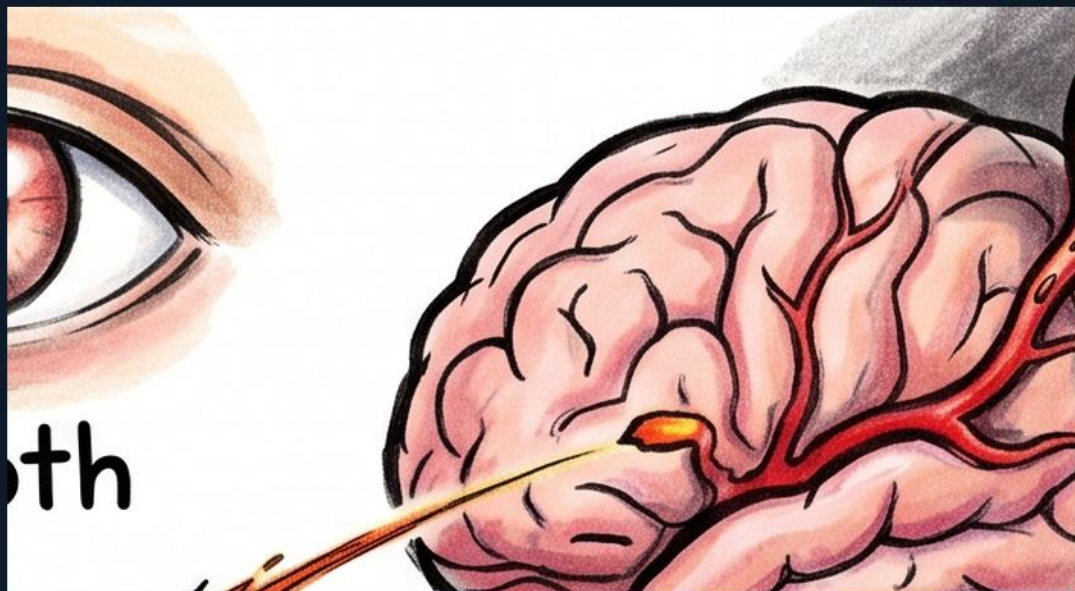
The eye with a pale-centered retinal hemorrhage labeled Roth is the ocular immunologic finding of IE.

WHAT IT ENCODES

Roth spots are retinal hemorrhages with pale/white centers (fibrin–platelet or inflammatory cells), an immunologic minor Duke criterion. They are NONSPECIFIC — also seen in leukemia, severe anemia, diabetes, hypertension, and HIV — so they support but do not prove IE.

THE BOARD TRAP

'Roth spots are diagnostic of endocarditis' -> wrong; Roth spots are nonspecific (leukemia, anemia, diabetes) and count only as ONE minor criterion — the diagnosis rests on the full Duke criteria with cultures and echo.



Brain stroke / mycotic aneurysm

WHAT YOU SEE

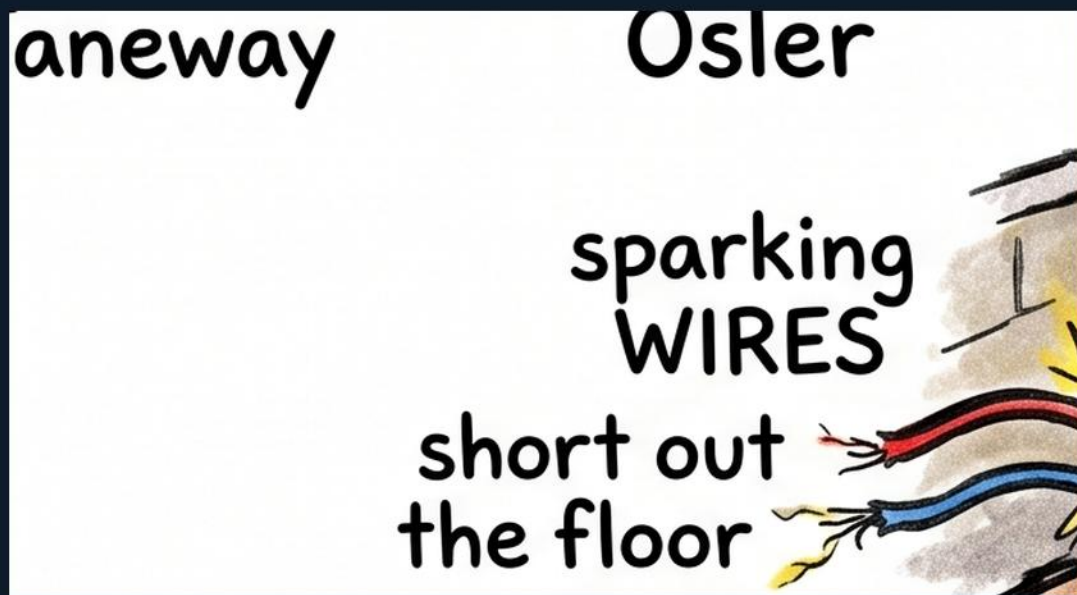
The brain struck by lightning with a bulging vessel is cerebral septic embolism and mycotic aneurysm — and the crossed-out hourglass warns against anticoagulation.

WHAT IT ENCODES

Septic emboli lodge in cerebral arteries (often MCA territory) causing embolic stroke; infection of the vessel wall forms a MYCOTIC ANEURYSM that can rupture into intracranial hemorrhage. CNS embolization is the most common extracardiac complication of left-sided IE and a major mortality source. After ischemic stroke without bleed, indicated valve surgery can proceed early; after intracranial hemorrhage, delay surgery ~3–4 weeks because bypass anticoagulation can expand the bleed.

THE BOARD TRAP

'Anticoagulate the IE patient who had an embolic stroke or has AF' -> wrong; anticoagulation in native-valve IE does NOT prevent embolization and risks converting a bland infarct into fatal hemorrhage (especially with an undiagnosed mycotic aneurysm) — the crossed-out hourglass/timer means do not rush to bypass after a bleed.



Aortic root abscess → AV block (sparking wires)

WHAT YOU SEE

The sparking electrical wires shorting out the floor under the throne, with an ECG, are aortic-root abscess invading the conduction system to cause heart block.

WHAT IT ENCODES

Perivalvular extension — most often from the AORTIC root, which abuts the AV node and His bundle — forms a periannular abscess that disrupts conduction. NEW PR prolongation, new bundle branch block, or new AV/heart block on ECG is the bedside red flag that infection has burrowed past the leaflet into the annulus. Get a TEE (~90% vs ~50% TTE for abscess) and monitor daily ECGs in aortic-valve IE.

THE BOARD TRAP

'New heart block in IE — continue IV antibiotics and recheck in a week' -> wrong; new conduction block means a periannular/aortic-root ABSCESS until proven otherwise and is a class I indication for urgent surgery — abscesses are not cured by antibiotics alone.



New murmur (trumpet)

WHAT YOU SEE

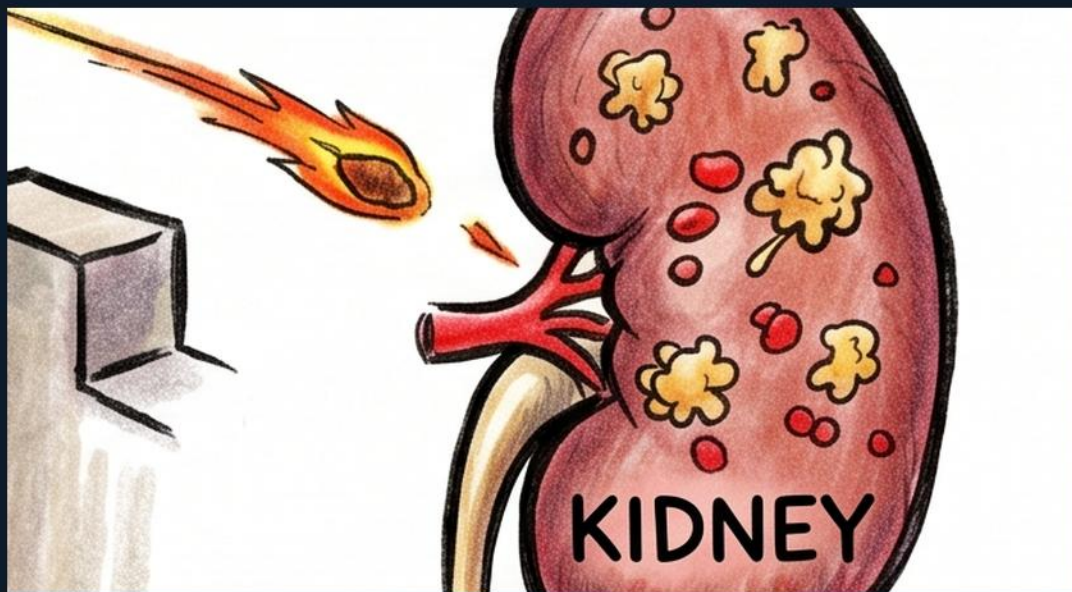
The trumpet blaring beside the throne is a new regurgitant murmur from valve destruction.

WHAT IT ENCODES

Valve leaflet destruction produces new/worsening regurgitation heard as a new murmur. A new murmur on auscultation alone is only supportive — it must be confirmed as NEW regurgitation on echo to meet the major imaging criterion. A new continuous machinery-like murmur with heart failure instead signals a fistula or ruptured sinus of Valsalva (a surgical emergency), not just worsening regurgitation.

THE BOARD TRAP

'A new murmur on physical exam is a major Duke criterion' -> wrong; new valvular regurgitation on ECHO is major, but a murmur heard on auscultation without echo confirmation does not meet the major bar — get the echo.



Immune-complex glomerulonephritis (kidney)

WHAT YOU SEE

The kidney studded with golden immune deposits labeled immune-complex glomerulonephritis is the renal immunologic complication.

WHAT IT ENCODES

Two renal mechanisms occur. Immune-complex deposition causes a diffuse proliferative GLOMERULONEPHRITIS (low complement, hematuria, RBC casts) — an immunologic phenomenon that RESOLVES with treating the infection. Separately, septic renal emboli cause flank pain/infarcts. Splenic septic emboli cause LUQ pain/infarcts and rarely a splenic abscess (persistent fever despite therapy warrants imaging).

THE BOARD TRAP

'Hematuria + low complement in an IE patient is a primary GN needing immunosuppression' -> wrong; in IE the glomerulonephritis is immune-complex mediated and resolves when the endocarditis is cured — the priority is curing the infection, not immunosuppression.

Summary — every symbol

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3. Osler nodes (painful pads)

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4. Splinter hemorrhages (fingertips)

Splinter hemorrhages are linear, reddish-brown subungual streaks caused by microemboli (or immune microvascular injury) to the nailbed capillaries, one of the peripheral stigmata of IE

5. Septic emboli to spleen/systemic

Left-sided vegetations embolize systemically

6. Roth spots (eye)

Roth spots are retinal hemorrhages with pale/white centers (fibrin-platelet or inflammatory cells), an immunologic minor Duke criterion

7. Brain stroke / mycotic aneurysm

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Perivalvular extension — most often from the AORTIC root, which abuts the AV node and His bundle — forms a periannular abscess that disrupts conduction

9. New murmur (trumpet)

Valve leaflet destruction produces new/worsening regurgitation heard as a new murmur

10. Immune-complex glomerulonephritis (kidney)

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